

8TH GRADE MATH • SUPPORT NOTES

Unit 7: Statistics

NC.8.SP.1-4

Name: _____

Date: _____

Class: _____

Period: _____

★ The One Big Idea

Data tells a story. We use **graphs and tables** to see patterns and make smart guesses.

□ Words to Know

Scatterplot — a graph of dots showing two things at once

Correlation — how two things move together

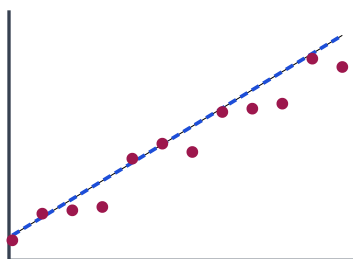
Outlier — a dot far away from the rest

Line of best fit — a straight line drawn through the middle of the dots

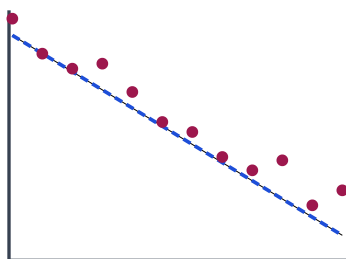
Trend — the overall direction of the data

7.1 Scatterplots & Correlation

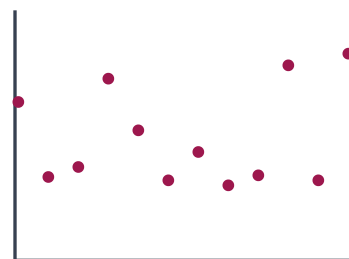
NC.8.SP.1



Positive (up ↗)



Negative (down ↘)



No correlation

Three Things to Describe

- **Direction:** positive (up), negative (down), or none.
- **Strength:** tight dots = strong; spread out = weak.
- **Outliers:** any dot that does not fit the pattern.

⚠ Correlation $\neq$ Causation

Two things moving together does **not** mean one **causes** the other. Ice cream sales and sunburns both go up in summer — ice cream does not cause sunburn!

7.2 Line of Best Fit

NC.8.SP.2, NC.8.SP.3

★ It Is Just $y = mx + b$

The line of best fit has a **slope** (rate) and a **y-intercept** (start), just like Unit 6.

Use it to **predict**: plug in an x , get a predicted y .

📈 Two Prediction Words

- **Interpolation**: predicting *inside* the data you have (safer).
- **Extrapolation**: predicting *outside* the data (riskier).

👤 I Do

Best fit line: $y = 2x + 5$ (x = hours studied, y = score). Predict a score for 6 hours: $y = 2(6) + 5 = 17$.

7.3 Two-Way Tables

NC.8.SP.4

 Reading the Table

	Likes Dogs	No Dogs	Total
Boys	12	8	20
Girls	15	5	20
Total	27	13	40

- **Joint:** one box (12 boys like dogs).
- **Marginal:** a total at the edge (20 boys, 27 like dogs).
- **Relative frequency:** a count $\div$ the total ($12/40 = 0.30 = 30\%$).

 You Try

What fraction of all students are girls who like dogs? $\frac{\quad}{40} = \quad\%$

Part 2 Practice **Answer**

1. A scatterplot of dots going downhill shows ____ correlation. _____
2. A single dot far from the rest is called an _____. _____
3. Best fit $y = 3x + 4$. Predict y when $x = 5$: _____
4. Predicting beyond your data is called _____. _____
5. From the table above, what % of students do NOT like dogs? _____ %

★ ANSWER KEY ★ **Practice**

1) negative 2) outlier 3) 19 4) extrapolation 5) $13/40 = 32.5\%$

You try (7.3): $15/40 = 37.5\%$.

End of Unit 7 support packet.